

Dementia and its treatments

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⚠ Q & A is open now - submit questions for me or for our guest



Slido Link

Outline

- 20 minutes on “Dementia”, to give a modern framework for understanding
- 20 minutes on the development of symptomatic treatments
- 5 minute break
- 20 minutes on “Disease modifying” treatments.
- 5 minutes for questions for me on the material presented
- 10-20 minutes Q and A trial participant with me
- 10 minutes Qs from students

Part 1: Dementia

A modern framework

Alzheimer's disease is not synonymous with dementia

Dementia is an umbrella term used to describe a state where an individual has lost the ability to carry out daily life activities independently, because of cognitive impairment.

There are many types and causes of dementia.

The most common ones are:

1. Alzheimer's disease (AD) **pathological definition**
2. Vascular dementia **aetiological definition**
3. Dementia with Lewy bodies **molecular definition**
4. Frontotemporal dementia **anatomical definition**

Nature Neuroscience overview

Inside Alzheimer's disease

nature video



Watch on

Question 1

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Alzheimer's disease is a disease like any other

- AD refers to the abnormal presence of A β and tau proteins which define AD among many other neurodegenerative diseases
- Some individuals may have dementia without A β or tau pathology, while others (rarely) may have no clinical symptoms, even in the presence of A β pathology.

Biological vs clinical definition

The old way pre-2010

- Clinical syndrome → “probable AD”
- Pathology only confirmed at post-mortem
- Heterogeneous cohorts in trials
- “*Dementia of the Alzheimer type*”

The modern way 2018 NIA-AA, 2024 revised criteria

- Biology defines the disease
 $A\beta$ + tau
- Clinical stage is described separately
- Biomarkers in life: CSF, plasma, PET
- Allows pre-symptomatic diagnosis

Question 2, Higher or Lower?

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The AT(N)(X) framework

A: Amyloid

T: Tau

N: Neurodegeneration

X: Other

What it measures: cerebral A β pathology.

- CSF A β 42/40 ratio **low = abnormal**
- Plasma A β 42/40 **replaces some lumbar punctures**
- Amyloid-PET **florbetapir, flutemetamol**



click to reveal: lumbar puncture (LP is painless)

Question 3

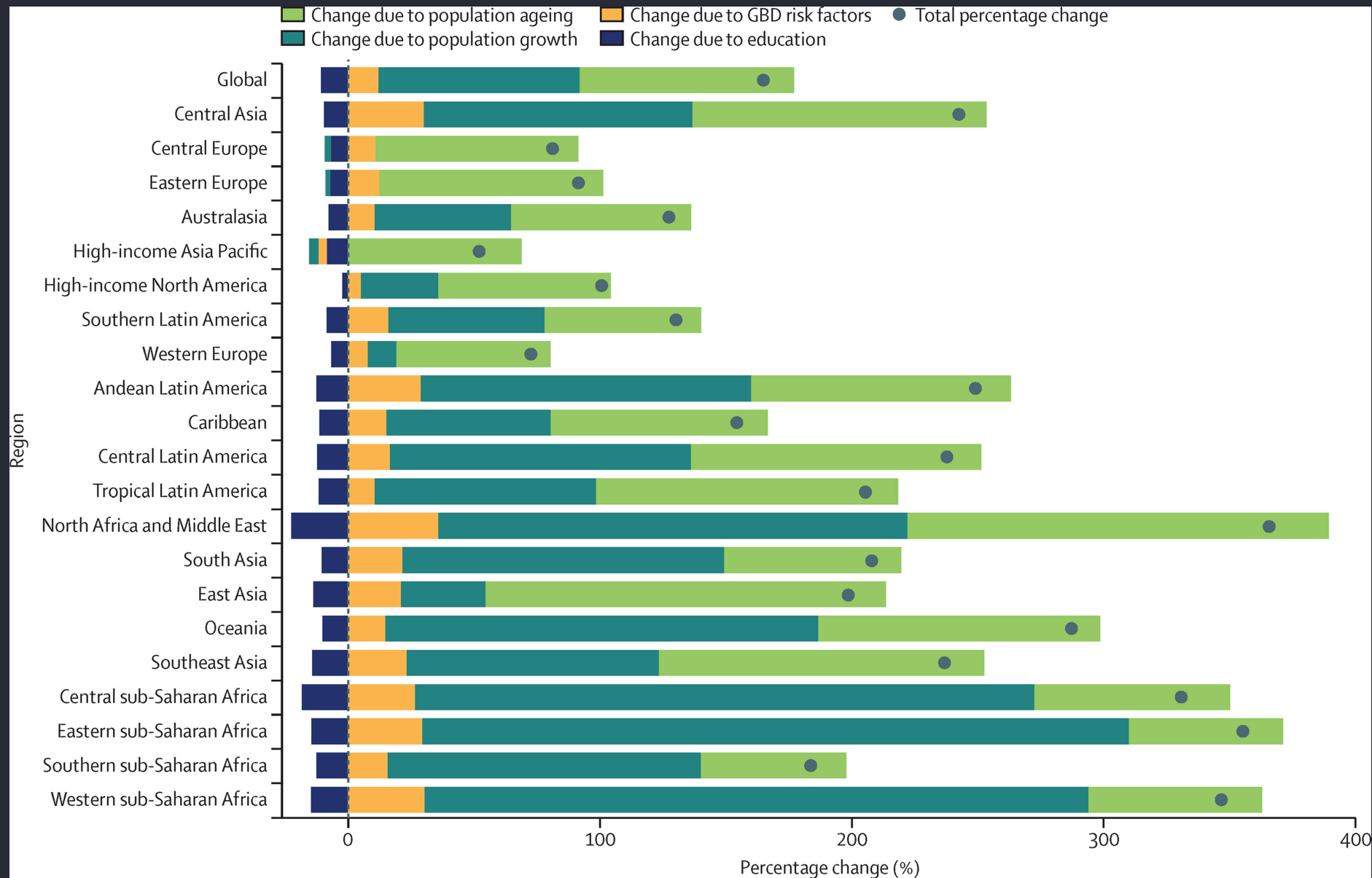
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The global prevalence of dementia is increasing

- There are many causes of dementia, but AD is the most common cause in individuals older than 65 years of age.
- In 2020, it was estimated that over 55 million people worldwide were living with neurodegenerative disease at the *dementia* stage.
- The number of people affected is set to rise to **152 million** by 2050, with the most significant increases in lower-middle-income (LMIC) countries.
- Dementia (including dementia due to AD) incidence and prevalence are increasing globally but are likely to be underestimated because of underdiagnosis and misdiagnosis
- As global lifespan continues to increase, the number of people affected by dementia is expected to rise. Up to three-quarters of those with dementia worldwide have not received a diagnosis.

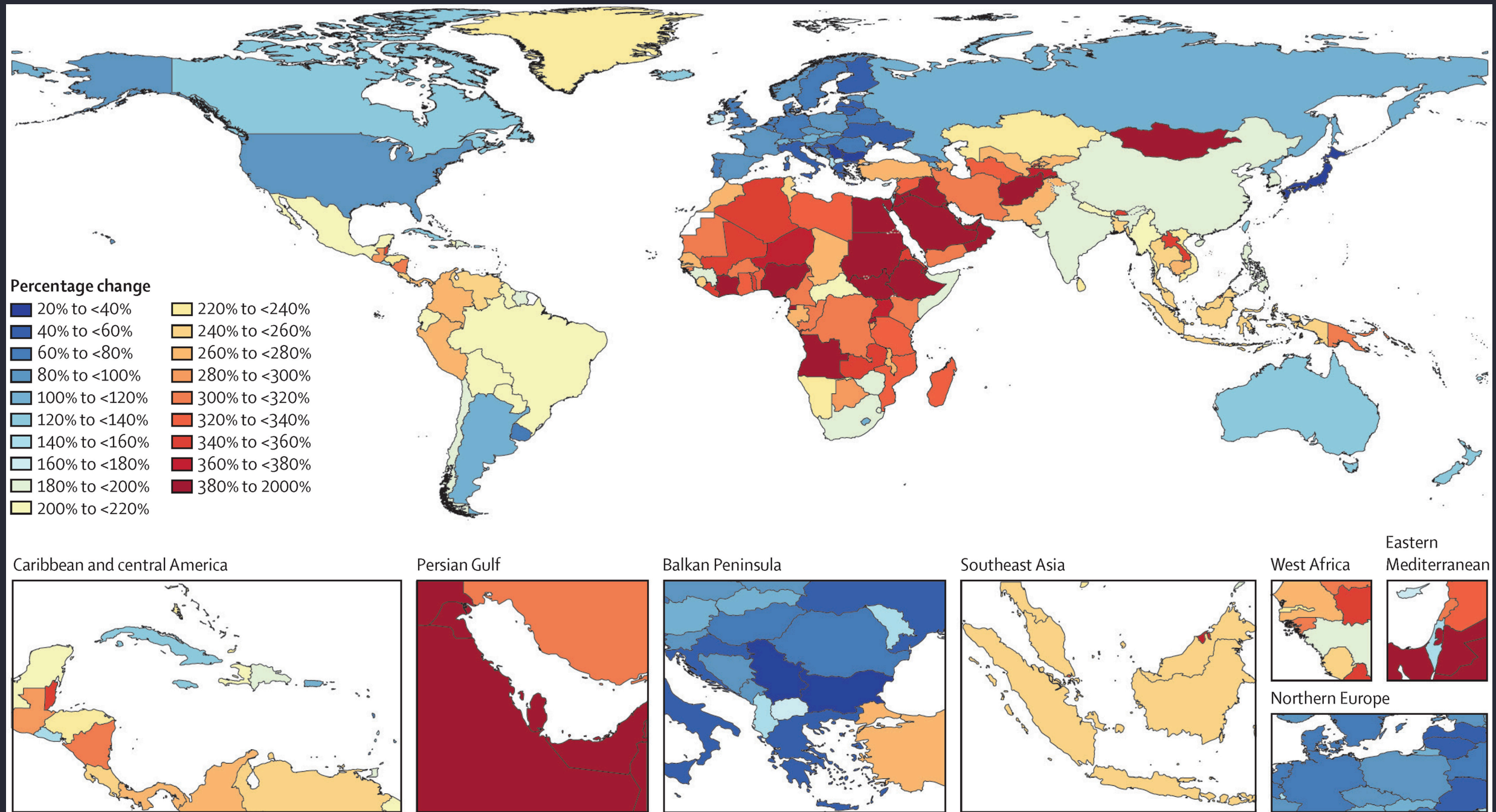
57.4M to 153M by 2050



Global change in dementia prevalence from: Figure 4 Decomposition of percentage change in the number of individuals with dementia between 2019 and 2050 globally and by world region

GBD=Global Burden of Diseases, Injuries, and Risk Factors Study; *Lancet PH* Volume 7, Issue 2e105-e125 February 2022

Regions of greatest change



Estimated change per region *Lancet PH* Volume 7, Issue 2e105-e125 February 2022

Historical development of Alzheimer's disease diagnostic criteria

- The 2023 revised criteria for diagnosis and staging of AD emphasizes that AD is a disease defined by its biology
- However, to avoid misdiagnosis, only biomarkers that have been proven to be accurate with respect to an accepted reference standard should be used for diagnosis
- It is important to note that biomarker tests should always be interpreted with clinical judgment

The clinician's expertise is crucial in the following situations:

1. When the biomarker test result contradicts the clinical presentation
2. When evaluating the probable contribution of AD versus other pathologies to clinical symptoms
3. When assessing how other medical conditions could possibly impact the biomarker results

Alzheimer's disease diagnostic biomarkers

Core 1 currently accepted for diagnosis

- CSF A β 42/40
- CSF p-tau181/A β 42
- CSF t-tau/A β 42
- Accurate plasma assays
- Amyloid-PET

Core 2 emerging / research-grade

- Plasma p-tau217 near-CSF accuracy
- Plasma MTBR-tau243
- GFAP astrocyte activation
- NfL non-specific neurodegeneration
- Tau-PET Braak staging in life

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Differential diagnosis

AD

DLB

FTD

VaD

- **Onset**: insidious, episodic memory first
- **Signature**: temporo-parietal cognition; language; getting lost
- **Imaging**: medial temporal and parietal atrophy
- **Treatment cue**: AChEi and memantine; now anti-amyloid eligible

Reversible mimics

Treatable contributors to cognitive impairment

- B12 / folate deficiency
- Hypothyroidism
- Depression *pseudo-dementia*
- Normal pressure hydrocephalus *gait, urinary, cognitive*
- Drug effects *anticholinergics, benzodiazepines, opioids*
- Delirium *often superimposed*
- Obstructive sleep apnoea
- Subdural haematoma
- Alcohol misuse

Assessment

Clinical History

Cognitive examination

Examination

Neuroimaging

Biomarkers

A medical history from the patient and collateral history from an appropriate informant.

- Educational attainment **quantitative and qualitative**
- Occupational attainment **manual/ skilled/ professional**
- Head Injury / Contact sports **rugby and boxing**
- Lifetime Alcohol + substances **peak, sustained**
- Smoking history
- Rapid forgetfulness
- Repetitive questioning
- Loss of recent memory **with relative preservation of remote**
- Functional decline **IADLs first, then ADLs**
- BPSD **agitation, depression, sleep**

Part 2: Symptomatic treatments

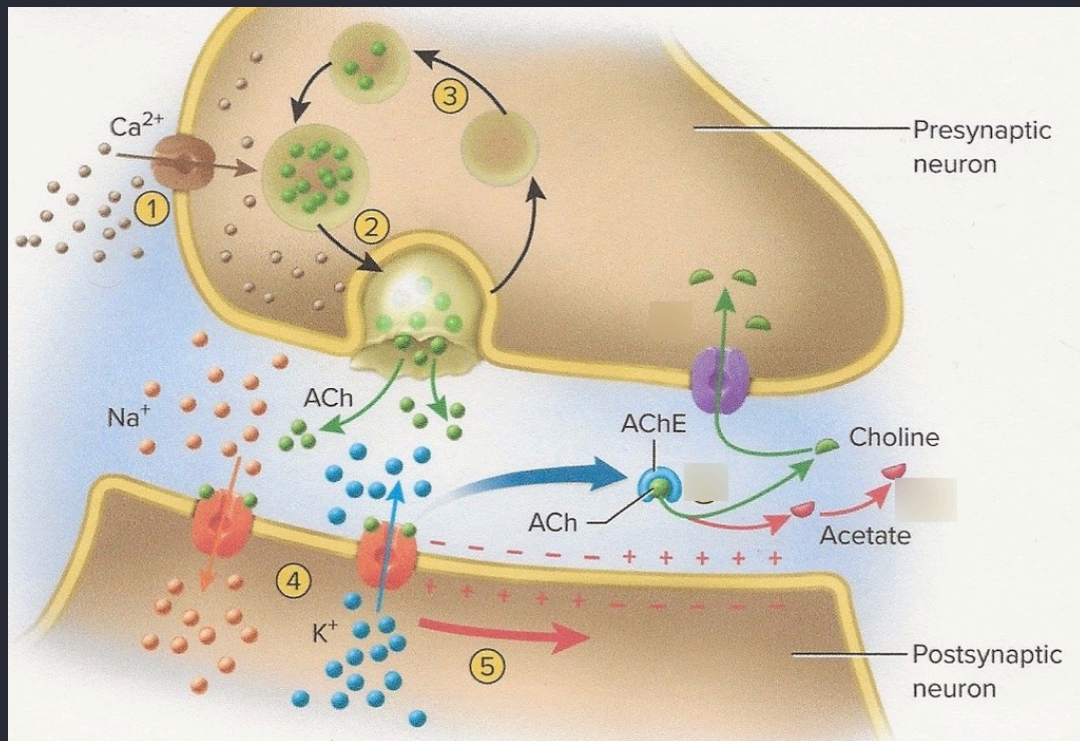
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The cholinergic hypothesis (1976)

- Bowen, Davies, Perry independently observed loss of choline acetyltransferase in post-mortem AD brain
- Source: degenerating cholinergic projections from the **nucleus basalis of Meynert**
- Implication: boost acetylcholine, improve cognition

AChE inhibitor mechanism



Cholinergic synapse

- AD brains have loss of cortical and hippocampal ACh
- AChEi inhibit synaptic breakdown of ACh, increasing transmitter availability per release
- Effect is presynaptic compensation; does not modify disease
- Side-effects from cholinergic excess: $\uparrow$ vagal tone, GI cramping, bradycardia, vivid dreams

AChE inhibitors: clinical use

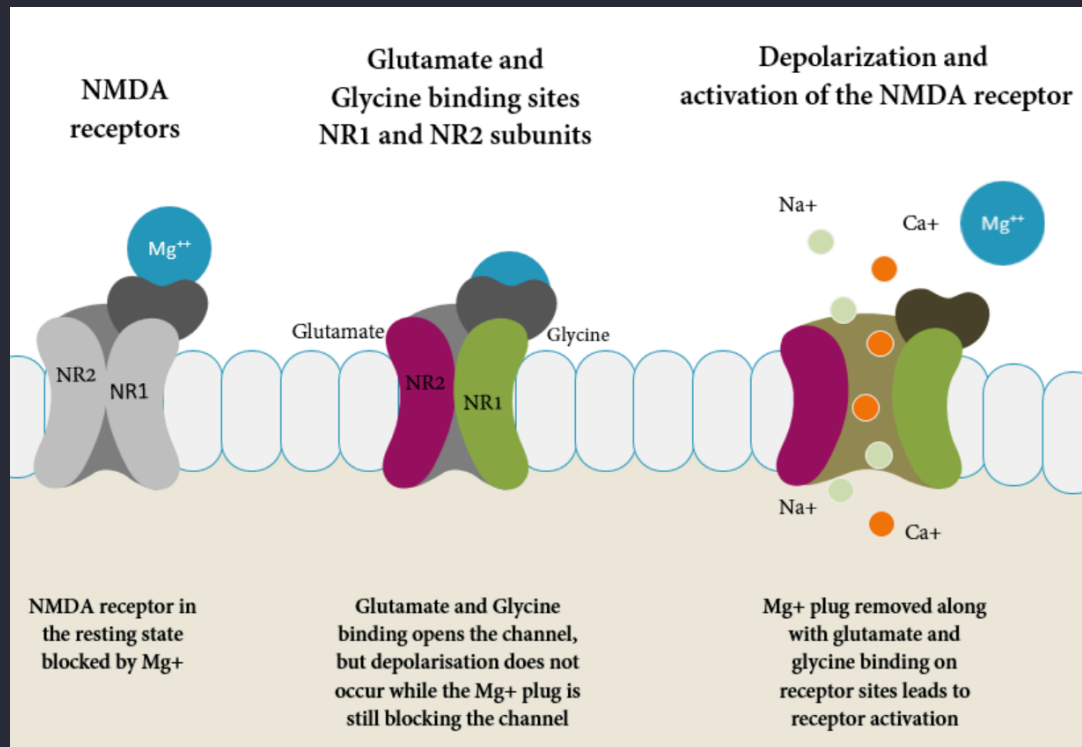
Donepezil

Rivastigmine

Galantamine

- Once daily, oral; 5 mg increased to 10 mg after 6 weeks **most common UK first-line**
- Long half-life
- Side effects: GI upset, vivid dreams, bradycardia
- Cardiac history → ECG before starting

Memantine: NMDA receptor antagonism



- Glutamatergic excitotoxicity → tonic NMDA-R activation → calcium overload
- Memantine sits in the channel only when it's pathologically open **voltage-dependent**
- Strong physiological NMDA bursts displace it, preserving physiological learning
- **Indication:** moderate to severe AD, or AChEi-intolerant
- Often combined with donepezil

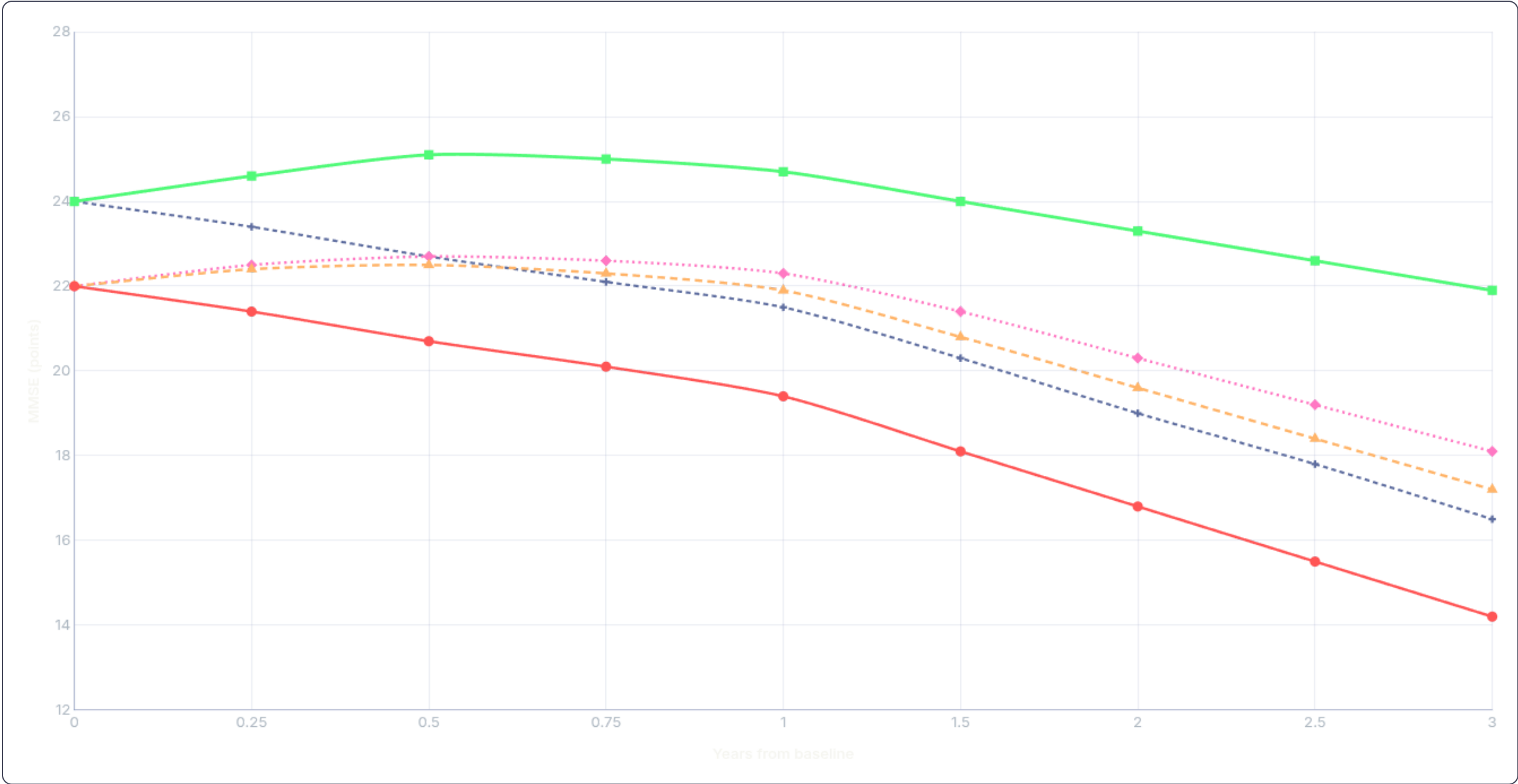
Memantine mechanism of action

Effect size of symptomatic treatment

MMSE trajectory by treatment and dementia type

Illustrative curves parameterised from Knight 2018, Xu 2021, Xu 2024, Truzzi 2022. Baseline MMSE 22 for AD, 24 for DLB.

Untreated AD AChEI in AD AChEI + memantine, moderate AD AChEI in DLB Untreated DLB



Curves on AChEI run roughly parallel to the untreated trajectory in AD, offset by ~1 MMSE point. The DLB response is meaningfully larger and more durable.

Differential responsiveness across subtypes

Helpful in

- **AD** modest, all severities
- **DLB** often substantial; rivastigmine first choice
- **PDD** cholinergic deficit prominent
- **VaD** mixed disease often benefits

Limited or harmful in

- **bvFTD** no benefit, may worsen agitation
- **Pure VaD** inconsistent
- **Severe AD** memantine takes over

BPSD

- **B**ehavioural and **P**sychological **S**ymptoms of **D**ementia
- Up to 90% of patients experience at least one over the course of illness
- Drives caregiver burden, institutionalisation, mortality
- Five clusters:
 1. Agitation / aggression
 2. Psychosis delusions, hallucinations
 3. Depression / anxiety
 4. Apathy
 5. Sleep disturbance

Non-pharmacological management of BPSD

- **Always rule out a precipitant:** pain, infection (UTI, chest), constipation, drug effect, sensory deprivation
- **Cognitive Stimulation Therapy** NICE-recommended, group-based
- **Exercise** aerobic + resistance
- **Music therapy:** strong evidence in agitation
- **Environmental modification:** light, routine, signage, familiar objects
- **Caregiver education and support:** often high-yield

Pharmacological management of BPSD

! Antipsychotics are last-resort

- ↑ stroke and all-cause mortality CATIE-AD, FDA black box
 - **Severe sensitivity in DLB**: can cause irreversible parkinsonism / NMS
 - Use only when behaviour endangers patient or others, lowest dose, short duration, regular review
-
- **SSRIs** for depression (citalopram has agitation evidence, but QTc concern)
 - **Brexpiprazole**: first FDA approval for agitation in AD, 2023 not yet NICE-approved on NHS for this indication
 - **Trazodone**: sleep, agitation; mostly off-label
 - **Avoid benzodiazepines**: falls, paradoxical agitation, dependence

Limits of symptomatic treatment

- They do not slow neuronal loss
- They do not clear A β or tau
- They do not change underlying trajectory
- A patient on AChEi 5 years from diagnosis has the same brain pathology as one untreated

What is important in clinical trials?

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5 minute break  

Part 3: Disease-modifying treatments

The amyloid hypothesis (Hardy & Higgins, 1992)

- **A β accumulation** is the upstream driver of AD
- Genetic anchor: every gene mutation that causes autosomal-dominant AD **APP**, **PSEN1**, **PSEN2** alters A β production or clearance
- Down syndrome (trisomy 21 $\rightarrow$ 3 copies of **APP**) $\rightarrow$ near-universal AD pathology by age 50
- Tau pathology, inflammation, neurodegeneration: downstream
- Predicted: lower A β $\rightarrow$ less downstream damage $\rightarrow$ better outcome

Failed anti-amyloid trials

- **Solanezumab**: Eli Lilly; failed Phase 3 in mild AD (2014, 2016)
- **Bapineuzumab**: Pfizer/J&J; failed (2012); high rate of vasogenic edema
- **Crenezumab**: Roche; failed (2019)
- **Gantenerumab**: Roche; first GRADUATE trials negative (2022); reformulated for prevention
- Lessons learned:
 - Treating too late **pathology has spread for 15+ years**
 - Targeting the wrong species **monomer vs oligomer vs plaque**
 - Insufficient brain penetration / dosing
 - Endpoint too noisy in mild populations

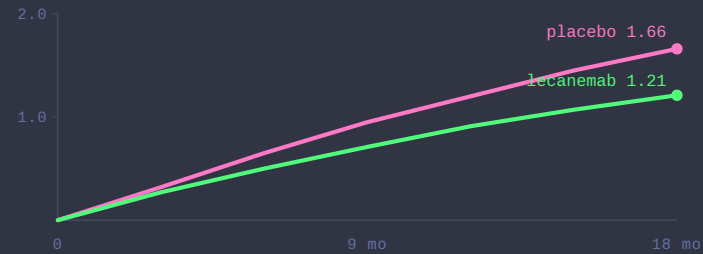
Aducanumab (2021)

- Two Phase 3 trials (EMERGE, ENGAGE): discordant; pre-specified analysis was negative
- Post-hoc reanalysis at high dose suggested cognitive benefit on EMERGE
- **FDA accelerated approval** June 2021, over near-unanimous advisory panel objection
- EMA **refused** authorisation
- NHS / NICE: **not commissioned**
- Eli Lilly withdrew it from market 2024

Lecanemab: CLARITY-AD (2023)

Phase 3 · $n = 1,795$ · 18 months · early symptomatic AD, biomarker-confirmed · target: soluble A β protofibrils

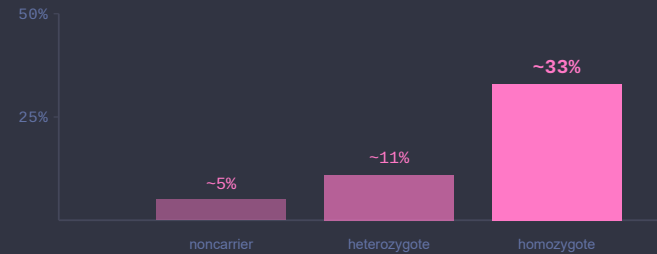
EFFICACY · CDR-SB CHANGE FROM BASELINE



27% slowing

-0.45 CDR-SB difference · $p < 0.001$ primary & all secondaries

SAFETY · ARIA-E RATE BY APOE GENOTYPE



12.6% / 17.3%

overall ARIA-E / ARIA-H · 9.2% symptomatic ARIA-E in homozygotes

REGULATORY

FDA · Jul 2023

full (traditional) approval

EMA · 2025

approved on appeal, excluding APOE4 homozygotes

NICE · rejected

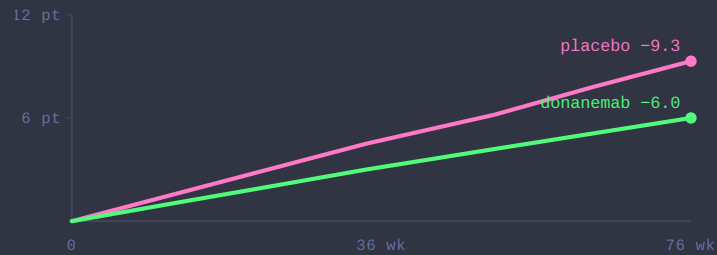
NHS use, Aug 2024 — not cost-effective at list price

van Dyck CH et al. *N Engl J Med*. 2023;388(1):9–21

Donanemab: TRAILBLAZER-ALZ 2 (2024)

Phase 3 · n = 1,736 · 76 weeks · MCI / mild AD · target: pyroglutamate-modified A β in plaques · stop-when-cleared design

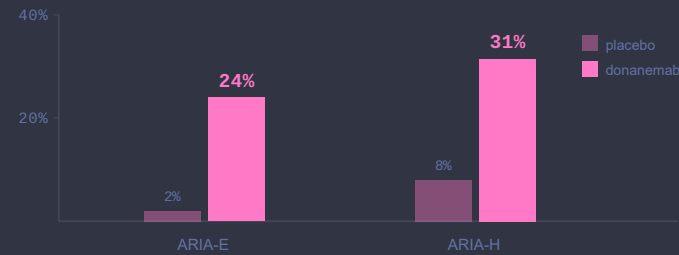
EFFICACY · IADRS DECLINE, LOW/INTERMEDIATE-TAU (N = 1,182)



35% slowing

iADRS · CDR-SB slowing 36% · 47% vs 29% no decline at 1 y

SAFETY · ARIA IN DONANEMAB ARM VS PLACEBO



5.8% / 1.5%

symptomatic / serious ARIA-E · 3 deaths attributed to ARIA

REGULATORY · APOE

FDA · Jul 2024

full approval

NICE · rejected

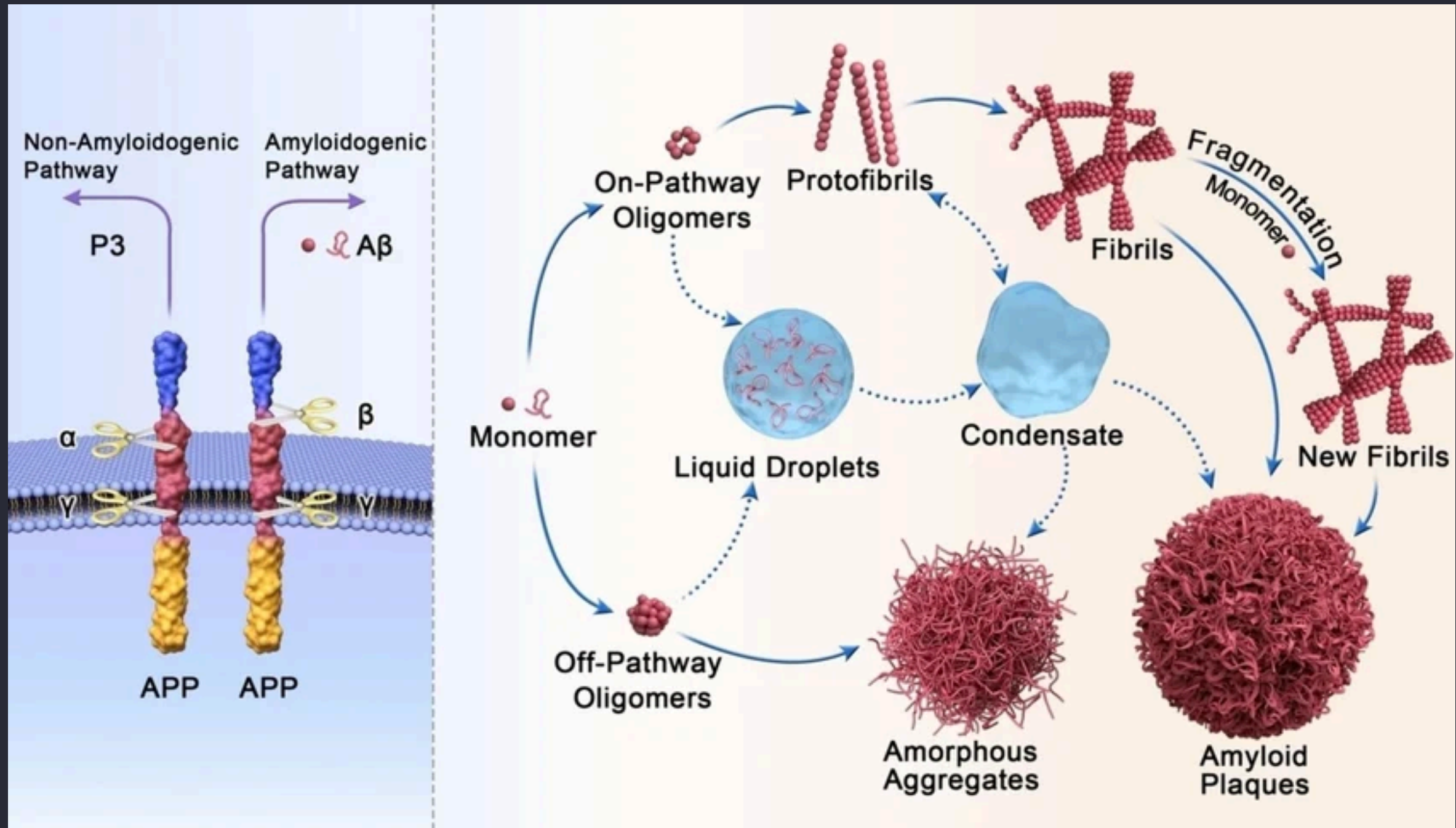
NHS use, Oct 2024

OR 4.57

ARIA-E in APOE4 homozygotes vs noncarriers

Sims JR et al. *JAMA*. 2023;330(6):512–527

Antibody specificity



Aβ aggregation pathway 10.1002/chem.202400277

Patient selection

- **Confirmed A β pathology**: amyloid-PET or CSF A β 42/40
- **Disease stage**: MCI or mild AD dementia (not moderate or severe)
- **MRI suitable**: no severe cerebral amyloid angiopathy, ≤ 4 microbleeds, no superficial siderosis
- **Anticoagulation**: caution or contraindication
- **APOE4 genotyping**: needed for ARIA risk counselling
- **Capable of attending infusions** every 2–4 weeks for 18+ months
- **Capacity** to consent

ARIA

ARIA-E oedema, effusion

- Vasogenic oedema, leptomeningeal effusion
- Best seen on **FLAIR**
- Most asymptomatic; caught on surveillance MRI
- Can present with headache, confusion, focal symptoms, seizures
- Severity is mostly mild; ~3% serious; rare deaths reported
- Surveillance MRI before infusions 5, 7, 14 (lecanemab) or before 2, 3, 4, 7 (donanemab)

ARIA-H haemorrhage

- Microbleeds, superficial siderosis
- Best seen on **SWI / GRE**
- Usually asymptomatic
- Persists indefinitely after resolution

APOE4 and ARIA risk

APOE genotype	Lecanemab ARIA-E	Donanemab ARIA-E
Non-carrier	~5%	~16%
Heterozygote	~11%	~23%
Homozygote	~33%	~41%

- ϵ 4 homozygotes are ~2% of the population but bear most of the risk
- **Counselling**: genotype results have implications beyond the individual family members, life insurance
- Donanemab US label restricts use in homozygotes; EMA mandates discussion of relative risk
- Patients can decline genotyping but must be counselled to the higher-risk assumption

UK implementation

- **Drug cost:** £20–25k per patient per year (lecanemab list)
- **Infrastructure:** infusion suite every 2 (lecanemab) or 4 (donanemab) weeks
- **Surveillance MRI:** 4–7 scans per patient course
- **Workforce:** trained nurses, dementia specialists, neuroradiologists, raters, pharmacists
- **NICE August 2024:** rejected lecanemab for NHS commissioning; benefit too small at price
- Currently available only privately in the UK; major equity concern
- Open question whether amyloid lowering alone is sufficient or whether combinations will be needed

Other approaches

- **Anti-tau antibodies:** E2814, JNJ-2056 (zagotenemab discontinued 2023) Phase 2
- **Tau ASOs** intrathecal antisense oligonucleotides, BIIB080
- **Inflammation:** TREM2 agonists in early trials
- **Vaccines:** UB-311, ABvac40 (Phase 2)
- **Prevention trials** in pre-symptomatic carriers:
 - **DIAN-TU:** autosomal-dominant AD families
 - **AHEAD 3-45:** sporadic, plasma p-tau217 stratified
 - *A4 closed early; solanezumab failed*

Current state

- The first disease-modifying drugs exist. Their effect size is modest. ARIA limits use. Access in the UK is restricted.
- **Diagnosis**: plasma p-tau217 may make primary-care testing feasible.
- **Prevention** may matter more at a population level than any antibody:
 - *Lancet Commission 2024: 14 modifiable risk factors account for ~45% of dementia risk*
 - Hearing, education, BP, smoking, obesity, depression, physical inactivity, diabetes, alcohol, social isolation, air pollution, head injury, vision, cholesterol

What is a Brain Health Centre?

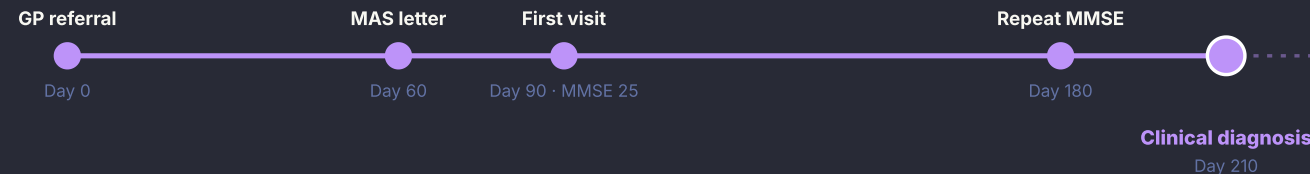
Same patient. Same disease. Different future.

A Brain Health Centre is not a relabelled memory clinic. It redesigns the pathway so that biomarkers, vascular workup, and behavioural support arrive together, early, before too much brain has been lost.

Mary, 64. "My memory's not what it was. My dad had dementia. Should I be worried?"

MEMORY ASSESSMENT SERVICE

cognition-led, no biomarkers, sequential appointments

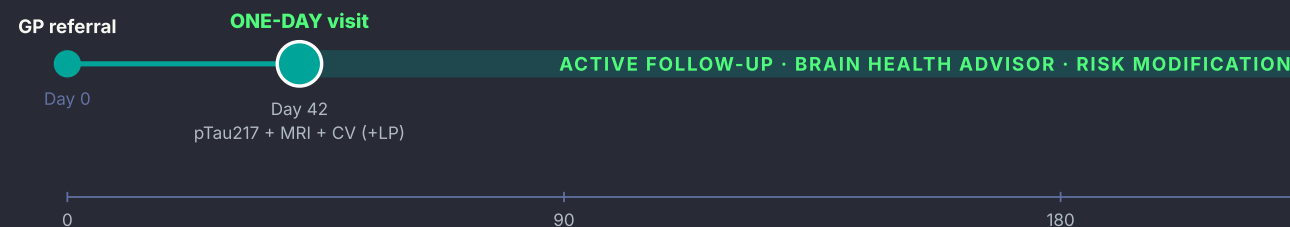


"probable Alzheimer's disease"

- × no biomarkers, no diagnostic certainty
- × ineligible for disease-modifying therapy
- × no structured risk-reduction plan

BRAIN HEALTH CENTRE

biomarker-led, integrated, with active follow-up



biomarker-confirmed diagnosis

- ✓ blood pressure normalised, statin started
- ✓ hearing aid referral, social plan
- ✓ DMT-ready, in research, year-on review booked

1 Confirm with biology, not exclusion. Plasma pTau217, MRI and CSF give a positive diagnosis of Alzheimer's pathology, instead of a diagnosis of last resort.

2 Treat the brain like the heart. Forty-five percent of dementia worldwide is linked to fourteen modifiable risk factors. Most are vascular and behavioural.

3 Be ready for therapy. Lecanemab and donanemab need confirmed amyloid. Without biomarkers, eligibility is impossible to establish.

Manchester Brain Health Centre

GMMH

mental health · old-age psychiatry · clinical lead

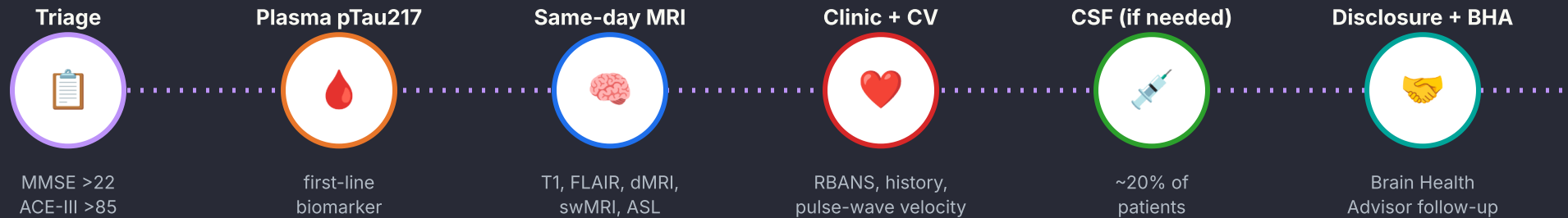
MFT

imaging · cardiology · labs · biobank host

academic partner: The University of Manchester

Alzheimer's Society

funding · Brain Health Advisors · public voice



ONE VISIT, ONE DAY (Phase 2, Q3 2026)

49_d

simulated median pathway, referral to disclosure

vs 187 days currently

73%

reduction in time to diagnosis

Phase 2 vs Phase 1b

~80%

of patients spared a lumbar puncture

blood-first triage

0

serious LP adverse events to date

~92% acceptance, atraumatic technique

Dementia - novel treatments

What's next, and why it matters

45% of dementia worldwide is linked to **14 modifiable risk factors** (Lancet Commission, 2024)

EARLY LIFE

Less education **5%**

MIDLIFE

Hearing loss **7%**

High LDL cholesterol **7%**

Depression **3%**

Traumatic brain injury **3%**

Physical inactivity **2%**

Smoking **2%**

Diabetes **2%**

Hypertension **2%**

Obesity **1%**

Excessive alcohol **1%**

LATE LIFE

Social isolation **5%**

Air pollution **3%**

Vision loss **2%**

What's next at MBHC

- One-stop blood-first pathway live in clinic from **Q3 2026**
- First UK site to deploy **pTau217 Elecsys** for routine NHS care
- Brain Health Advisors trained in **motivational interviewing**, embedded in pathway
- Five-year cohort study with **GSK**, ASL imaging endpoints for future trials
- Hub-and-spoke model across **Greater Manchester**

Why it matters for psychology

- Pathology starts **20+ years** before symptoms. Behaviour change has the most leverage when started early.
- Disease-modifying drugs (lecanemab, donanemab) are MHRA-licensed but **not yet on the NHS**. Eligibility depends on biomarkers.
- Most modifiable risk factors are **psychological and behavioural**: hearing aid use, exercise, sleep, social engagement, mood.
- This is where psychologists work, with clinicians, on the things that actually shift the curve.

Questions on the material (5 min)



Q&A with trial participant (10–20 min)

Welcome!



Student questions (10 min)

Summary

1. **Dementia is a syndrome; AD is defined biologically.** $A\beta$ and tau can be detected in life via CSF, plasma p-tau217, or PET.
2. **Symptomatic treatments help modestly and do not change disease course.** AChEi and memantine offer about a 6-month delay. Non-pharmacological BPSD interventions are first-line. Antipsychotics carry mortality risk.
3. **Anti-amyloid antibodies are the first disease-modifying drugs.** Effect size is modest. ARIA limits use. NICE has rejected them for the NHS. Prevention may matter more.

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